

## PROBLEM STATEMENT

Logarithmic Properties &amp; Chain Rule

Find the derivative with respect to  $x$  of the function:

$$y = \log \left( \frac{x+1}{x-1} \right)$$

STEP 1 প্রদত্ত লগারিদমীয় মূলদ ফাংশনটি উল্লেখ করি (Given Function)

$$y = \log \left( \frac{x+1}{x-1} \right)$$

STEP 2 লগারিদমের ভাগ সূত্র প্রয়োগ করে সরল আকারে বিশ্লেষণ করি (Applying Logarithmic Quotient Law)

$$y = \log(x+1) - \log(x-1)$$

STEP 3 উভয়পক্ষে  $x$ -এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করি (Applying Differentiation Operator w.r.t.  $x$ )

$$\frac{dy}{dx} = \frac{d}{dx} [\log(x+1)] - \frac{d}{dx} [\log(x-1)]$$

STEP 4 চেইন রুল প্রয়োগ করে প্রতিটি পদের পৃথক অন্তরজ নির্ণয় করি (Differentiating Each Term via Chain Rule)

$$\frac{dy}{dx} = \frac{1}{x+1} \cdot \frac{d}{dx}(x+1) - \frac{1}{x-1} \cdot \frac{d}{dx}(x-1) = \frac{1}{x+1}(1) - \frac{1}{x-1}(1) = \frac{1}{x+1} - \frac{1}{x-1}$$

STEP 5 বীজগাণিতিক ভগ্নাংশের লসাগু নিয়ে চূড়ান্ত সরল মান পাই (Simplifying by Common Denominator)

$$\frac{dy}{dx} = \frac{(x-1) - (x+1)}{(x+1)(x-1)} = \frac{x-1-x-1}{x^2-1} = -\frac{2}{x^2-1}$$

**FINAL ANSWER**  $\frac{dy}{dx} = -\frac{2}{x^2-1} \quad \left( \text{or, } \frac{2}{1-x^2} \right)$

**BOX 1: MATHEMATICAL FOUNDATIONS & FORMULA REFERENCE**

- **Logarithmic Quotient Property:**  $\log\left(\frac{u}{v}\right) = \log u - \log v$ , converting division into simple subtraction prior to differentiation.
- **Standard Derivative of Logarithm:**  $\frac{d}{dx}[\log u] = \frac{1}{u} \frac{du}{dx}$ . Here  $\frac{d}{dx}[\log(x \pm 1)] = \frac{1}{x \pm 1} \cdot 1$ .
- **Difference of Squares:**  $(x + 1)(x - 1) = x^2 - 1$ .

**BOX 2: COMMON STUDENT TRAPS & PITFALLS**

- **Numerator Subtraction Sign Error:** When combining fractions,  $(x - 1) - (x + 1) = x - 1 - x - 1 = -2$ . A common mistake is writing  $x - 1 - x + 1 = 0$ .
- **Unnecessary Heavy Chain Rule:** Differentiating directly without expanding logarithms leads to  $\frac{1}{\frac{x+1}{x-1}} \cdot \frac{d}{dx}\left[\frac{x+1}{x-1}\right]$ , which is algebraically much more prone to errors under exam pressure.

**BOX 3: REAL DOMAIN & DIFFERENTIABILITY ANALYSIS**

- **Real Domain:** Natural log requires strictly positive arguments:

$$\frac{x+1}{x-1} > 0 \iff (x+1)(x-1) > 0 \iff x \in (-\infty, -1) \cup (1, \infty)$$

- The function is undefined in the closed interval  $[-1, 1]$ .
- **Differentiability:** Infinitely smooth and differentiable everywhere on  $(-\infty, -1) \cup (1, \infty)$ .

**BOX 4: ALTERNATIVE DIRECT METHOD (CHAIN RULE + QUOTIENT RULE)**

- Let  $u = \frac{x+1}{x-1}$ , then  $y = \log u$ .

$$\frac{du}{dx} = \frac{(x-1) \cdot 1 - (x+1) \cdot 1}{(x-1)^2} = \frac{-2}{(x-1)^2}$$

- Applying Chain Rule:

$$\frac{dy}{dx} = \frac{1}{u} \cdot \frac{du}{dx} = \frac{x-1}{x+1} \cdot \frac{-2}{(x-1)^2} = \frac{-2}{(x+1)(x-1)} = -\frac{2}{x^2-1}$$

**BOX 5: EXAM MARKS BREAKDOWN (AMIE SECTION-A FORMAT)**

- **Step 1–2 (Log Expansion):** 1.0 Mark
- **Step 3–4 (Term-by-Term Differentiation):** 1.5 Marks
- **Step 5 & Final Answer (Algebraic simplification):** 1.5 Marks
- **Total Allocation: 4.0 Marks**